

BidLens AI: Project Specification & AI Handover Blueprint

Smart India Hackathon 2026 | Problem Statement ID: SIH26100
AI-Powered Integrated Bid Compliance & Value Intelligence Platform for GeM Procurement

Problem ID: SIH26100	Domain: Public Procurement / GeM / AI	Status: Specification & Roadmap Locked
Target Contest: Smart India Hackathon	Feasibility Rating: 9.0 / 10 (Student MVP)	National Procurement Volume: █4+ Lakh Crore

1. Executive Problem Definition & Core Pain Points

The Operational Challenge: Government e-Marketplace (GeM) processes over █4 Lakh Crore worth of public procurement annually. For a single tender, government evaluation officers receive 10 to 50 vendor bid proposals. Each proposal is an 80 to 150-page PDF packed with legal clauses, financial balance sheets, technical specifications, OEM authorizations, and statutory certificates. Manually reading thousands of PDF pages under strict administrative deadlines is mentally exhausting, resulting in costly human evaluation errors, inadvertent disqualification of compliant MSMEs, and frequent court litigation by aggrieved bidders.

The 'Familiar Brand Bias' Dilemma: Because manual review is so taxing, officers subconsciously default to large, famous brand names they have previously purchased from. Small startups and local MSMEs that offer superior technical specifications, 5-year warranties (vs 1 year standard), or 24/7 dedicated onsite support at lower prices get overlooked simply because their bonus advantages are buried deep on Page 74 of their proposals.

Project Mission (BidLens AI): To deliver an AI-assisted procurement intelligence platform that automates strict rule compliance checking, maps clauses into a traceable Knowledge Graph, cross-checks documents for contradictions, flags rejection risks with evidence citations, and actively highlights 'Value-for-Money' bonus perks offered by smaller vendors.

2. The 5 Core Innovation Pillars (Teacher-Aligned Novelties)

- Pillar 1: Clause-to-Evidence Compliance Knowledge Graph:** Constructs a directed semantic Knowledge Graph mapping: *Regulatory Rules (GFR 2017 / MSME Orders) → RFP Bid Clauses → Required Evidence → Submitted Vendor Attachments → Verified Compliance Decisions*. This replaces opaque AI scores with end-to-end audit traceability.
- Pillar 2: Cross-Document Contradiction Detection Engine:** Performs cross-attachment entity linking and consistency checks across GSTIN certificates, PAN cards, audited financial statements, OEM Manufacturer Authorization Forms (MAF), and self-declarations to detect legal entity name mismatches, signatory discrepancies, or conflicting turnover figures.
- Pillar 3: Explainable Rejection-Risk Engine:** Identifies likely technical and financial disqualification points before final bid award. Delivers evidence-grounded reasons with exact PDF page citations and regulatory clause references rather than an uninterpretable black-box score.
- Pillar 4: Regulatory Change-Aware Knowledge Base:** Continuously monitors and models changes in GeM General Financial Rules (GFR), Public Procurement (Make in India) Orders, and MSME statutory policies, dynamically updating the compliance validation rules without code rewrites.
- Pillar 5: Bid Repair & Re-Verification Assistant:** Recommends specific corrective remediation actions for detected non-compliance gaps (e.g. *'Upload Annexure IV with active GSTIN verification'*), accepts revised draft submissions, and automatically rechecks the updated proposal in seconds.

3. Full-Stack Technology Architecture & Tool Ecosystem

Layer / Domain	Open-Source Tools & Libraries	Functional Role in Prototype
Frontend UI	Streamlit / React.js + TailwindCSS	Interactive officer dashboard, drag-and-drop PDF uploader, side-by-side comparison tables, and visual graph explorer.
Backend API	Python FastAPI / Flask, Pydantic	Asynchronous REST API managing multi-PDF uploads, background processing queues, and structured JSON scoring schemas.
Document & OCR	PyMuPDF, pdfplumber, EasyOCR, Tesseract	Extracts text, table grids, and font metadata from 150-page PDFs; OCR cleans scanned certificates, seals, and stamped letters.
Knowledge Graph	Neo4j Community / NetworkX	Builds the directed Clause-to-Evidence Knowledge Graph linking regulations, bid requirements, and submitted evidence nodes.
RAG & Embeddings	ChromaDB / FAISS, LangChain, BGE-M3	Vector database semantically indexing document chunks for sub-second clause retrieval and equivalence matching.
AI Reasoning	Ollama (Mistral-7B) / Gemini API Free Tier / Groq	Grounded LLM reasoning for clause verification, evidence citation extraction, and MSME bonus perk summarization.
Security & Audit	hashlib (SHA-256), ReportLab, Pandas	Generates file checksums for anti-tampering, pre-scans prompt injection, and exports certified audit dossiers (PDF/Excel).

4. Real-World Edge Cases & Anti-Jailbreak Defenses

- **Sneaky Fine Print Exclusions:** Vendors claiming '3-year warranty' on Page 5 but excluding display/battery on Page 80. The AI triggers conditional NLP pattern matchers, flagging 'Conditional Compliance' for officer review.
- **MSME Statutory Exemptions:** Indian Public Procurement Policy for MSEs (Order 2012) grants registered MSMEs exemption from prior turnover and experience. System checks Udyam certificates first to prevent unfair disqualifications.
- **Functional Equivalence:** Tenders specifying Intel i7 (16-core) matching AMD Ryzen equivalents via semantic vector embeddings rather than naive keyword matching.
- **Cross-Document Inconsistencies:** Detects conflicting values across attachments (e.g. 30-day delivery in cover letter vs 60 days in technical annexure; name mismatch between GSTIN and OEM MAF).
- **Data vs Instruction Sandboxing:** PDF text is encapsulated inside `` XML blocks. System prompt forces LLM to treat document text strictly as passive evidence, blocking indirect prompt injection attacks.
- **Invisible Text / White Font Scrubber:** PyMuPDF inspects font color, opacity, and size (<2pt) to strip white-on-white text inserted by dishonest vendors attempting to game automated keyword matchers.
- **Deterministic Code Rules:** Hard facts (turnover numbers, expiry dates, EMD amounts) are calculated via Python math rules, not LLM probabilistic inference, guaranteeing 100% mathematical precision.

5. Government Security Architecture, Data Usage & Budget ROI

Security / Practical Dimension	Architecture Specification & Real-World Impact
Sovereign On-Premise AI	Sensitive national defence and railway tender data NEVER leaves the private government perimeter. Runs on local servers or NIC MeghRaj Cloud with zero public cloud data retention.
SHA-256 Checksum Fingerprinting	Every uploaded PDF is mathematically fingerprinted instantly. If an insider or hacker modifies a single character or price later, the hash mismatch triggers an immediate lockdown alert.
Blind Technical Evaluation	Automatically redacts vendor logos and company names during technical evaluation (naming them Vendor A, Vendor B) to completely eliminate big-brand favoritism.
Remote Area / Offline Edge Mode	100% offline capability: PDF parsing, OCR, and local rule matching run locally on an officer's laptop consuming 0 KB data . When online, consumes < 100 KB JSON tokens per tender.
National Expenditure & ROI	Entire nationwide deployment costs ~₹25L to ₹45L/year (~₹2 to ₹3 per tender). By catching just 0.01% of tender fraud or high-value MSME deals, it saves the government ₹40 to ₹100+ Crore annually (1000x ROI) .

6. 7-Stage Step-by-Step Implementation Roadmap

- **Phase 1: Dataset & Groundwork:** Curate real GeM tender RFPs, GFR procurement rules, and synthesize 3 diverse test bids: MegaTech (Big Brand), Apex Labs (High-Perk MSME), and Global Corp (Ineligible vendor with expired certificates).
- **Phase 2: PDF & OCR Ingestion Pipeline:** Build PyMuPDF text and table extraction pipeline + EasyOCR for scanned certificates, ink stamps, and OEM Manufacturer Authorization Forms (MAF).
- **Phase 3: Knowledge Graph Construction:** Construct Clause-to-Evidence Compliance Knowledge Graph (Neo4j / NetworkX) semantically linking GFR rules → RFP clauses → submitted PDF evidence nodes.
- **Phase 4: Contradiction & MSME Value Engine:** Develop cross-document entity contradiction detector for GST/PAN/turnover figures + the 'Value-for-Money' MSME bonus perk spotlight algorithm.
- **Phase 5: Rejection-Risk Engine & Bid Repair:** Build Explainable Rejection-Risk engine, Bid Repair Assistant for corrective re-verification, and multi-layer prompt injection / white-font sanitizers.
- **Phase 6: Full-Stack Web Application:** Develop responsive Streamlit / React officer dashboard featuring interactive Knowledge Graph visualizers, side-by-side comparison matrices, and 1-click certified PDF/Excel audit export.
- **Phase 7: Offline Edge & Demo Polish:** Package 100% offline edge mode + create 1-click desktop launcher (Start_BidLens.bat) and rehearse the 5-minute SIH live jury demonstration.

7. AI Handover & Instant Execution Prompt for Any New AI Session

Instructions for any AI Agent receiving this PDF on a new laptop / account:

- You are tasked with building the working prototype for **BidLens AI (SIH26100)**. All specifications, tool stacks, edge cases, and feature requirements are fully detailed in this document. When the user says 'Build the BidLens app', execute the following steps immediately without asking unnecessary clarifying questions:
1. Create the project structure with `app.py` (Streamlit frontend/dashboard), `pdf\_engine.py` (PyMuPDF & EasyOCR parser), `graph\_engine.py` (Knowledge Graph & RAG clause matcher), and `security.py` (SHA-256 & prompt sanitizers).
 2. Generate synthetic sample PDFs for testing: `Tender\_RFP.pdf`, `Bid\_VendorA\_BigBrand.pdf`, `Bid\_VendorB\_MSME\_HighValue.pdf` (offering 5-yr warranty & 32GB RAM upgrade), and `Bid\_VendorC\_Ineligible.pdf` (with expired GSTIN & contradictory PAN).
 3. Provide a single-click `Start\_BidLens.bat` desktop launcher so the user can double-click and run the live app in their browser.